

In short

In **offline reinforcement learning**, key challenges are **limited sample size** and **state-action distribution mismatch** between the *target (candidate) policy* and *behavioral (logging) policy*. In this work, we propose SimuDICE, an approach that uses Dual stationary Distribution Correction Estimation (DualDICE [1]) along with an estimation of model prediction confidence to improve the sampling of synthetic experiences. This approach improves the policy by balancing experiences similar to those frequently encountered with ones that have a distribution mismatch. SimuDICE achieves performance comparable to existing algorithms while requiring fewer pre-collected experiences and planning steps. Moreover, the algorithm remains robust across varying data collection policies. These improvements suggest a potential reduction in the hallucination risk by lowering the need for extensive sampling.

Background

Reinforcement Learning (RL): An agent uses a policy to interact with its environment, aiming to maximize the cumulative expected return.

Offline RL: optimizes policies using statically pre-collected datasets, which is essential when collecting new data is costly or risky. The challenge lies in learning from limited datasets, $\mathcal{D} = \{(s_0^{(i)}, s^{(i)}, a^{(i)}, r^{(i)}, s'^{(i)})\}_{i=1}^N$, which might have been generated by multiple, often sub-optimal, policies.

Model-based RL and Planning: Enables expansion beyond the limited dataset by generating synthetic experiences. Planning, using a trained world model, predicts future states to improve the policy performance without direct interactions. Figure 1 shows Dyna-Q [2], one of the first frameworks to unify learning, planning, and acting in *online RL*.

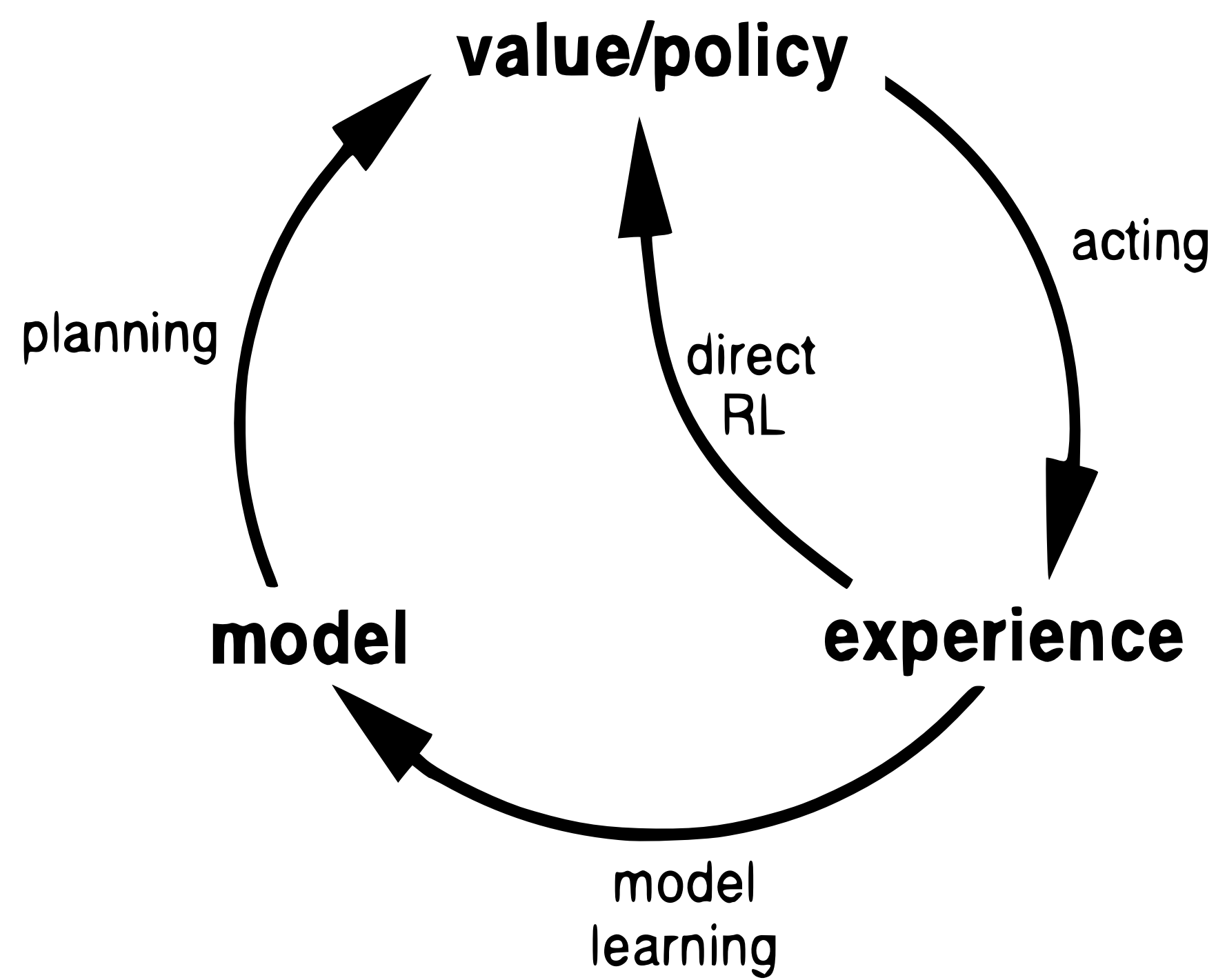


Figure 1. Relationships among learning, planning, and acting. Figure taken from Dyna-Q [2].

Distribution Correction Estimation (DICE): This technique estimates for each transition in the offline dataset how the reward shifts compared to the expected reward that would result from running a particular policy in the same environment.

DualDICE estimation: DualDICE [1] achieved impressive results for DICE estimation by reducing the problem to density estimation and using a change-of-variable technique. Eq. (1) shows how policy evaluation is transformed into density ratio estimation, while Eq. (2) the estimated weights.

$$\rho(\pi) = \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\frac{d^\pi(s,a)}{d^{\mathcal{D}}(s,a)} r(s,a) \right] \quad (1)$$

$$w_{\pi/\mathcal{D}}(s,a) := \frac{d^\pi(s,a)}{d^{\mathcal{D}}(s,a)} \quad (2)$$

SimuDICE

We begin by adapting Dyna-Q [2] to the offline setting (Figure 2). In this approach, we use offline data to (1) pre-train a one-step forward dynamic model of the environment - a Tabular World Model, and (2) learn an initial Q-value policy. This framework then generates synthetic experiences by sampling from the model, improving on the initial policy, leading to the *target policy*.

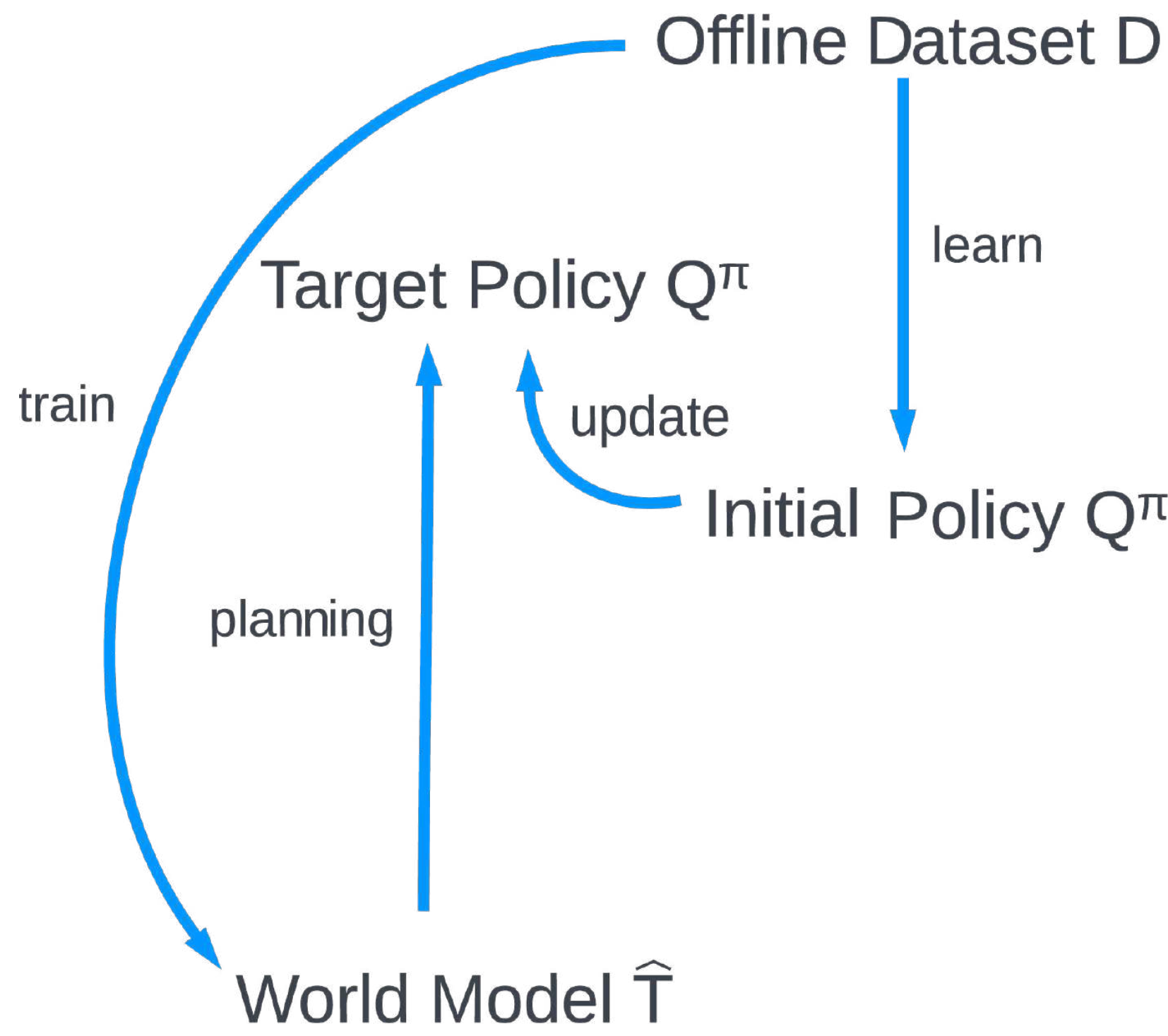


Figure 2. The offline adaptation of Dyna-Q [2].

World Model: We used a Tabular World Model for simplicity in this work. This memory-based model replicates previously observed experiences by selecting the most frequently occurring next state and averaging rewards, being able to manage stochastic environments.

SimuDICE: We propose a way to steer the synthetic experiences generation towards similar experiences to those frequently encountered, as well as experiences with a distribution mismatch. Moreover, SimuDICE can iterate until convergence, as shown in Figure 3.

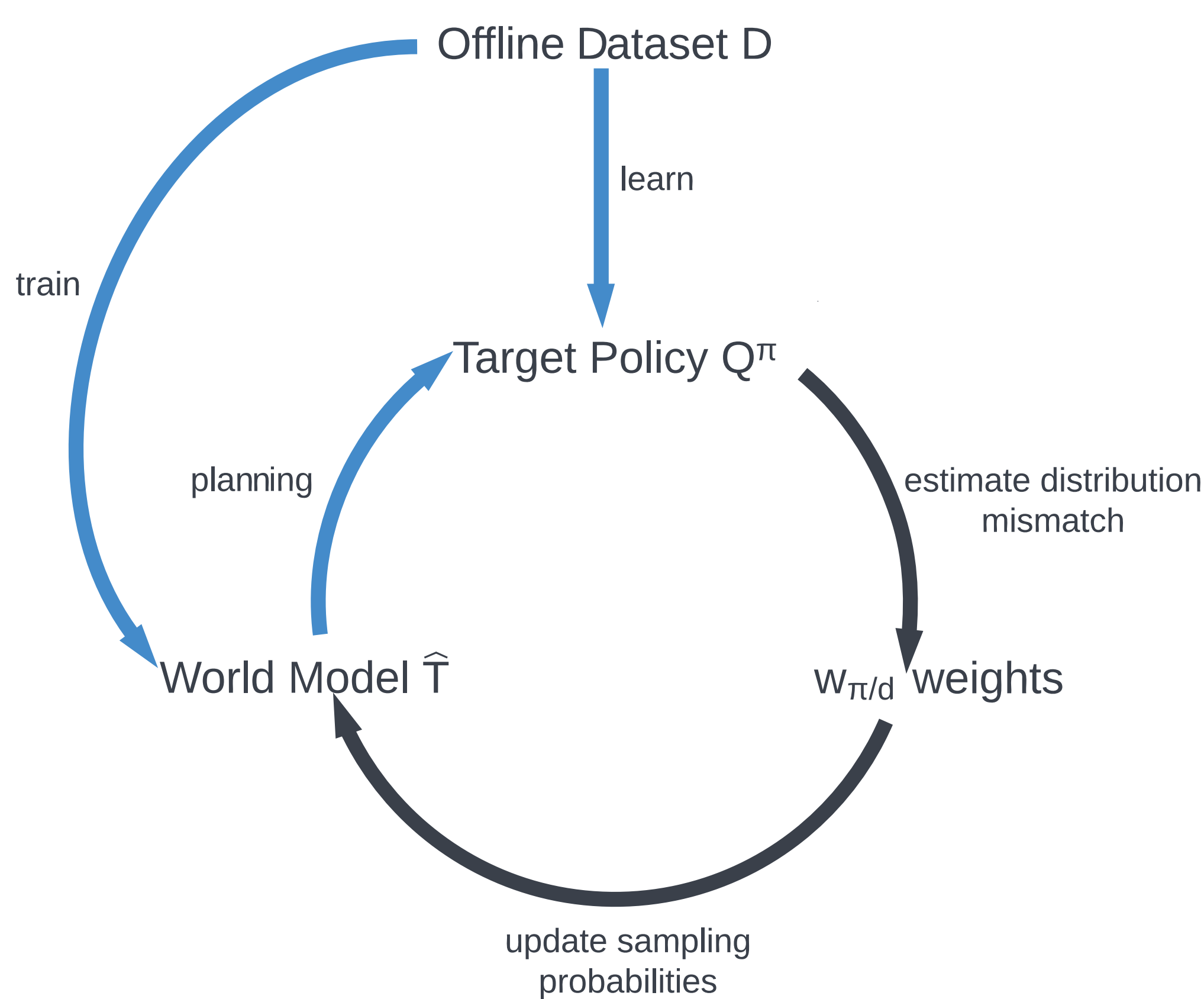


Figure 3. The components of SimuDICE and their interactions. Transitions adapted from Dyna-Q [2] are in blue, while those unique to SimuDICE are depicted in black.

Distribution mismatch estimation: Is achieved through DICE estimation, for which we reimplemented the $w_{\pi/\mathcal{D}}$ weight estimation from DualDICE [1].

Model confidence estimation: We estimated the confidence of a prediction $\mathcal{C}(s,a)$ as the normalized frequency of occurrences in the offline dataset.

Sampling probability updates: The sampling updates are performed by summing the model's confidence estimation with the λ -regularized softmax of the $w_{\pi/\mathcal{D}}$ weights (Eq. 3), followed by normalization (Eq. 4).

$$\mathcal{L}(s,a) = \mathcal{C}(s,a) + \frac{e^{w_{\pi/\mathcal{D}}(s,a) \cdot \lambda}}{\sum_{(s',a')} e^{w_{\pi/\mathcal{D}}(s',a') \cdot \lambda}} / \lambda \quad (3)$$

$$\mathcal{P}(s,a) = \frac{\mathcal{L}(s,a)}{\sum_{(s',a')} \mathcal{L}(s',a')} \quad (4)$$

Experiments

Figure 4 compares SimuDICE with two other algorithms: offline Q-learning and the offline adapted Dyna-Q [2] with uniform sampling probabilities.

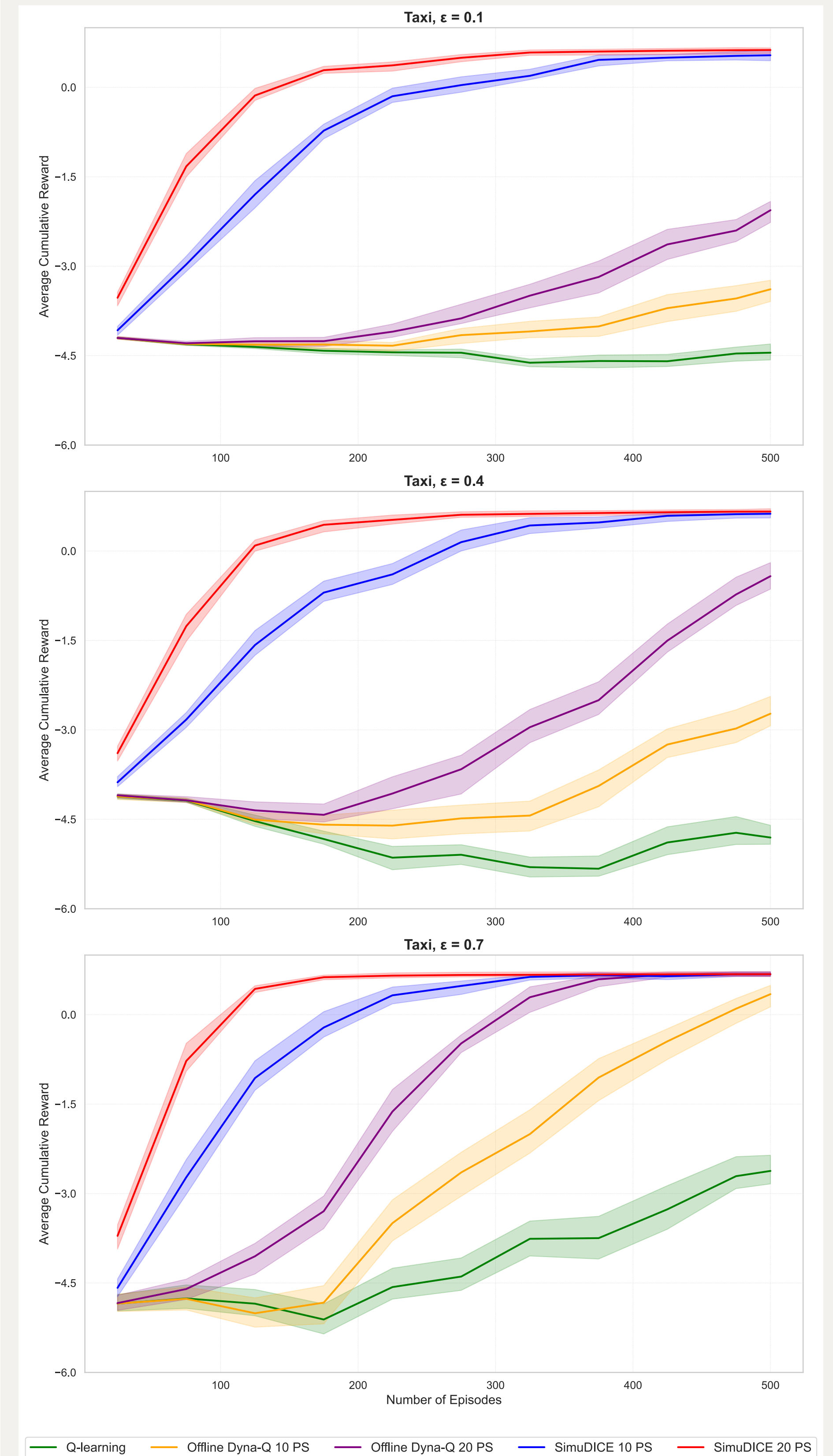


Figure 4. Comparison of algorithm performance in discrete tabular environments: *Taxi*, *CliffWalking*, and *FrozenLake*, under varying epsilon-greedy data collection policies ($\epsilon = 0.1, 0.4, 0.7$). Each plot shows the average cumulative reward as a function of the number of trajectories in the offline data. The results are averaged over 500 episodes and 5 different random seeds. The shaded regions represent the variance across the different seeds. PS represents the number of planning steps.

In the *Taxi* environment, which is considered more challenging due to the larger state-action space and more diverse actions, SimuDICE significantly outperforms the other algorithms. SimuDICE with only 10 planning steps outperforms the offline Dyna-Q with 20 planning steps and shows greater robustness across the different data collection policies.

Takeaways

SimuDICE achieves comparable results to similar offline RL algorithms while requiring **less data** and **fewer simulated experiences**, and it is robust to different data collection policies. It improves sample efficiency and reduces the impact of distribution mismatch by adjusting sampling probabilities using DualDICE [1] and prediction confidence estimates.

However, testing was limited to a few simple environments and used basic algorithm components.

Future work should explore more complex settings with continuous state-action spaces and world models capable of generating novel experiences.

References

- [1] O. e. a. Nachum, "DualDICE: Behavior-agnostic estimation of discounted stationary distribution corrections," in *Advances in Neural Information Processing Systems*, 2019, pp. 2315–2325.
- [2] R. S. Sutton, "Dyna, an integrated architecture for learning, planning, and reacting," *ACM SIGART Bulletin*, vol. 2, no. 4, pp. 160–163, 1991.